ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 4

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 style (design-related, 80 marks). Section A: objective questions (20) · Section B: designing a regulated and solar power supply for a robot, sensor and driver circuits with prototyping and fault finding, and logic design with a PLC ladder rung for a sorting conveyor (60).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw circuit diagrams neatly, using the correct symbols. A ruler and a non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The integrated circuit most often used to give a fixed 5 V output in a robot power supply is the …

A 555. B 7805. C 741. D 2N2222.

1.1.2 The component that smooths the ripple after a rectifier is a …

A capacitor. B fuse. C LDR. D relay.

1.1.3 A bridge rectifier uses … diodes.

A one B two C three D four

1.1.4 When the light falling on an LDR increases, its resistance …

A increases. B stays the same. C becomes infinite. D decreases.

1.1.5 A flywheel diode across a relay coil protects the transistor against …

A overheating of the LED. B high-voltage spikes when the coil switches off.

C short circuits on the board. D reverse polarity of the AC mains.

1.1.6 The correct order of a PLC scan cycle is …

A update outputs, run program, read inputs. B run program, read inputs, update outputs.

C read inputs, run program, update outputs. D read inputs, update outputs, run program.

1.1.7 The two "universal" logic gates are …

A AND and OR. B XOR and XNOR.

C NAND and NOR. D NOT and AND.

1.1.8 Identical solar panels connected in series give a …

A higher voltage. B higher current only.
C lower voltage. D lower power.

1.1.9 The five holes in one short row of a breadboard's terminal strip are …

A connected to each other. B insulated from each other.
C connected to the power rails. D connected to every other row.

1.1.10 The tool used to remove melted solder from a joint is a …

A side cutter. B wire stripper.
C long-nose pliers. D desoldering pump.

(10)

1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.2.1–1.2.10).

	COLUMN A		**COLUMN B**
1.2.1	Changes AC into pulsating DC	A	comparator
1.2.2	Keeps the output voltage constant when the load changes	B	soldering
1.2.3	Small AC variation left on a DC output	C	rectifier
1.2.4	Sensor whose resistance falls as the temperature rises	D	truth table
1.2.5	Limits the current through an LED	E	NTC thermistor
1.2.6	Op-amp circuit that switches its output when one input is higher than the other	F	solenoid
1.2.7	Electromagnet that moves a plunger in a straight line	G	ripple
1.2.8	Shows the output for every combination of inputs	H	ladder logic
1.2.9	Graphical PLC language that looks like a relay circuit	I	voltage regulator
1.2.10	Joining components with a molten metal alloy	J	series resistor

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 2: POWER SUPPLY DESIGN FOR A ROBOT

2.1 A robot controller needs a steady 5 V DC supply made from the 230 V AC mains. Draw a neat, labelled block diagram of a regulated DC power supply and state the function of EACH block. (5)

2.2 Explain why a bridge rectifier is preferred to a half-wave rectifier in this power supply. (2)

2.3 Explain how the filter capacitor reduces ripple. (2)

2.4 A simple Zener regulator uses a 5,1 V Zener diode fed from a 9 V supply. The load draws 20 mA and the Zener must carry 10 mA. Calculate:

2.4.1 The value of the series resistor (3)

2.4.2 The power dissipated in the series resistor (2)

2.5 The transformer must deliver 12 V to a load that draws 1,5 A. Calculate the minimum VA rating of the transformer. (2)

2.6 For outdoor work the robot's 12 V, 20 Ah battery is charged by two solar panels, each rated 18 V and 3 A, connected in parallel.

2.6.1 Calculate the output voltage, current and power of the panel group. (2)

2.6.2 The charge controller delivers 5 A to the battery. Calculate how long it takes to charge a completely flat battery, ignoring losses. (2)

[20]

QUESTION 3: SENSOR AND DRIVER CIRCUITS

3.1 A light sensor for a line-following robot uses an LDR in series with a 10 kΩ resistor across a 5 V supply. The output is taken across the 10 kΩ resistor. The LDR measures 100 kΩ in the dark and 1 kΩ in bright light.

3.1.1 Calculate the output voltage in the dark and in bright light. (4)

3.1.2 State whether the output voltage rises or falls when the light gets brighter. (1)

3.2 A red LED (2 V, 20 mA) is driven from a 5 V microcontroller pin. Calculate the value of the current-limiting resistor and its power dissipation. (3)

3.3 Draw a neat, labelled circuit diagram in which a microcontroller output pin switches a 12 V relay through a base resistor and an NPN transistor. Show the flywheel diode and the 12 V supply. (6)

3.4 The relay circuit was moved from the breadboard to Veroboard, but now it does not work. Name FOUR checks you would do, with or without a multimeter, to find the fault. (4)

3.5 Explain how an op-amp used as a comparator can switch on a fan when a temperature sensor's voltage rises above a set reference voltage. (2)

[20]

QUESTION 4: LOGIC DESIGN FOR A SORTING CONVEYOR

4.1 A sorting conveyor has three sensors: A = 1 when a part is present, B = 1 when the part is metal and C = 1 when the part is tall. The reject arm Y must push a part off (Y = 1) when a part is present AND it is either NOT metal OR it is tall.

4.1.1 Draw the truth table for Y. (4)

4.1.2 Write the Boolean expression for Y. (2)

4.1.3 Draw the logic circuit for Y using NOT, OR and AND gates. (4)

4.2 The same control is to be done by a PLC. Draw the ladder-logic rung for Y, using contacts for A, B and C and an output coil Y. (4)

4.3 Show how a NOT gate can be made from a single NAND gate, and explain why NAND is called a universal gate. (2)

4.4 The reject arm is driven by a solenoid. Explain how a solenoid produces a straight-line push. (2)

4.5 Give TWO ways to make the solenoid push harder. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF QUESTION PAPER